# Distinguished Lecture Series II: Shing-Tung Yau, "The Basic Tools to Construct Geometric Structures"

Terence Tao · 17 May 2007

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On Thursday, Yau continued his lecture series on geometric structures, focusing a bit more on the tools and philosophy that goes into actually building these structures. Much of the philosophy, in its full generality, is still rather vague and not properly formalised, but is nevertheless supported by a large number of rigorously worked out examples and results in special cases. A dominant theme in this talk was the interaction between geometry and physics, in particular general relativity and string theory.

As usual, there are likely to be some inaccuracies in my presentation of Yau's talk (I am not really an expert in this subject), and corrections are welcome. Yau's slides for this talk are available here.

In Yau's first lecture, Yau informally defined a "geometric structure" as an object (such as a metric or a special coordinate system) which induced a special connection which had the additional properties of being torsion-free and having a special holonomy group (e.g. SU(n) for Calabi-Yau manifolds, Sp(n) for hyperkähler manifolds, etc; Yau remarked that the exceptional Lie group $G_2$, the spin group Spin(7), and the unitary groups U(n) also seemed to play particularly important roles). But in this talk, Yau refined his previous assertion, saying that more recently, it has been understood that this metric, connection, and holonomy structure often needs to be enriched with additional structures (fibrations (or more generally, families of submanifolds), moduli space structure, special cycles, etc.) in order to fully capture the geometry of the situation, and in particular to shed light on some (still poorly understood) phenomena in geometry and physics, such as mirror symmetry. It is probably no coincidence that these types of additional structures also make a frequent appearance in string theory.

For instance, some manifolds are fortunate enough to enjoy an interpretation as a moduli space of classes of some other type of geometric object; this certainly doesn't occur for all manifolds, but when it does, it seems that this interpretation is fundamentally important to understanding the geometry of the manifold. In Zhang's lecture two weeks ago, we saw that the upper half plane (modulo a lattice) could be viewed as a moduli space of elliptic curves (or tori); more generally, Yau recalled that the Siegel upper half-space could be interpreted as the moduli space of abelian varieties. Somewhat remarkably, many of these moduli spaces naturally come with complex, Hermitian, Kähler, or hyperkähler structure. For instance, many hyperkähler manifolds can themselves be viewed as moduli spaces of semistable bundles over other hyperkähler manifolds.

In order to exploit this moduli space structure properly, there is a general and fundamental construction that generates nonlinear (and highly non-trivial) maps from one geometric object to another geometric object (which may be very different in topology, structure, or dimension); this construction seems to play a deep (and not fully understood) role in the analysis of such structures. A model example of this construction comes when trying to map structures (such as cohomology classes or sheaves) from one symmetric space G/H of a Lie group G to another symmetric space G/K. Both of these symmetric spaces are a projection of a common space $G/(H \cap K)$. If one has some universal object on that common space, such as a canonical bundle B, then in some cases, one can canonically convert a structure on the first space G/H by pulling it up to the common space, twisting it somehow by the universal object (e.g.

for cohomology classes, one might multiply through by some canonical class associated to this object, such as a Chern class or Todd class), and then pushing back down to G/K.

$$\begin{array}{ccccc} & & B & & \\ & & \downarrow & & \\ & & G/(H \cap K) & & \\ & \swarrow & & \searrow & \\ G/H & & & & G/K \end{array}$$

As observed by Chern, this general construction can be used to explain many classical constructions and identities, such as the kinematic formulae in integral geometry of Poincaré, Santalo, and Blaschke (in which the incidence relation between points and lines (say) is itself viewed as a symmetric space of the relevant group of motions, and serves as the common space mentioned earlier, projecting down to the space of points and the space of lines separately). (Yau also briefly mentioned the Hecke correspondence in number theory as another classical example of this construction, but didn't give details.) But it also can be used to interpret more modern constructions. For instance the Donaldson polynomial invariants of a four-manifold M can be viewed in terms of this construction, by introducing the moduli space $\mathcal{M}$ of rank 2 vector bundles over M with self-dual curvature; the product space $M \times \mathcal{M}$ then acquires a tautological rank two bundle, the second Chern class of which can be used to twist cohomology as we pull back from M and push forward onto $\mathcal{M}$.

Another important example of a phenomenon which can be understood as a special case of this fundamental construction is that of T-duality from string theory. In its simplest form, T-duality starts with a torus $T = \mathbb{R}^n/\Lambda$ and then considers its dual torus $T^* := (\mathbb{R}^n)^*/\Lambda^*$, where $\Lambda^*$ is the dual lattice to $\Lambda$ (note that the Pontryagin dual of T is $\Lambda^*$, not $T^*$!). For instance, in one dimension, the dual of a circle of radius r is essentially the circle of radius 1/r. The dual torus has a natural interpretation as the moduli space of complex flat line bundles over T (indeed, the holonomies of any such bundle are a multiplicative homomorphism from $\Lambda$ to U(1), and thus can be identified with an element of the Pontryagin dual of $\Lambda$, which is $T^*$). This interpretation gives rise to a tautological complex line bundle L over the product $T \times T^*$. The exponent of the first Chern class of this line bundle (which lives in the even-dimensional cohomology) can then be used to twist cohomology classes as they are pulled back from T and pushed forward onto $T^*$; this nonlinear transformation between the cohomology of T and the cohomology of $T^*$ underlies the T-duality phenomenon in string theory.

As in other areas of mathematics, this concept becomes even more powerful when one considers *families* of such constructions, rather than just an individual construction. For instance, Strominger, Yau, and Zaslow showed that algebraic Calabi-Yau manifolds M of complex dimension 3 (thus real dimension 6) can be fibred (outside of a singular set of high codimension) as a 3-torus bundle over the 3-sphere $S^3$. Applying the T-duality construction to all the tori in this bundle, one obtains a new algebraic Calabi-Yau manifold $M^*$ (after closing up the singularities). If one performs some additional transforms on the base sphere (basically a Legendre transformation on an affine structure on this sphere – which Yau described in his previous talk – taking into account some monodromy issues arising from the singularities), one then creates a non-trivial transform from geometric structures on M to those on $M^*$. This transform (which seems to have initially been motivated by brane theory – a quantised version of string theory) appears to be a good candidate for explaining the (still mysterious) mirror symmetry phenomenon arising from conformal field theory; there is a good match arising from computing specific examples and in comparing partial results known about both operations. Even without a complete understanding of this correspondence, though, these ideas have already been used to obtain nontrivial new results, for instance it was used by Candelas et al. to discover a formula to count algebraic curves in Calabi-Yau manifolds, which could be rigorously established using the works of Liu-Lian-Yau and Givental.

In this setting, the T-duality transform maps holomorphic bundles over M (or more precisely, a combination of the Chern and Todd classes of that bundle, which is essentially the Mukai vector of that bundle) to minimal Lagrangian cycles on $M^*$ (or more precisely, their representatives in $H^3(M^*)$), thus connecting even-dimensional cohomology of one manifold to the odd-dimensional cohomology of its

dual. Given that both sides of this correspondence consist of purely algebraic objects, Yau proposed that something similar should happen in arithmetic geometry: in particular, the action of K-groups defined by algebraic vector bundles on M should somehow correspond to the Frobenius map in etale cohomology of $H^3(M^*)$. Unfortunately, as Yau remarked, the methods coming from physics seem ill-equipped to handle finite characteristic, and so it is not clear how to formalise such correspondences at all.

The T-duality construction should be generalisable to other contexts, in which the torus T is replaced with some other nice space, and the dual $T^*$ replaced with some suitable moduli space of bundles over that space; Yau mentioned some recent progress in this direction in understanding manifolds with holonomy group $G_2$. In this case, the torus T gets replaced with a K3 surface.

The existence of special fibrations in a manifold can be used to construct other types of geometric structures. For instance, it is an open problem to construct an explicit metric on a K3 surface whose holonomy group is SU(2). Using ideas from physics, Greene, Shapere, Vafa and Yau used a K3 surface which can be fibred (outside of a singular set) as a torus bundle of flat tori over a two-sphere to almost solve this problem; the only remaining issue with the metric they constructed is to remove the singularity. From physical considerations, this metric should be a semiclassical approximation of a family of smoother metrics related to the quantum theory of the K3 surface, which should be expressible as a formal power series expansion involving sums over holomorphic disks (which have a physical interpretation as instantons), but apparently this intuition has not yet been made fully rigorous. Mirror symmetry also seems to connect these disks to holomorphic cycles on the dual manifold, though I didn't understand this point very well. It is also of interest to find out what the analogue of this story is in one higher dimension, for three-dimensional Calabi-Yau manifolds; this seems to be more difficult because the tori involved have three real dimensions and thus have no complex structure. (Yau remarked that complex structures are "given by God", whereas when you only have real structure, you have to work a lot harder to reach a comparable grasp of the manifold.)

We have seen how manifestly geometric objects such as bundles, fibrations and cycles allow us to understand the geometry of functions. But it is also useful to employ more analytic objects, such as the solutions of various geometric differential equations (or the geometric differential operators themselves) to also achieve understanding (somewhat in analogy to how algebraic varieties can be understood through their ring of functions). A good example is Donaldson's theory that analyses the topology of 4-manifolds through the moduli space of solutions to the Seiberg-Witten equations or the (somewhat similar) self-dual Yang-Mills equations. [Yau remarked that the former can be viewed as a sort of deformation of a supersymmetric version of the latter.] Yau seemed to feel that we were still far from achieving the full potential of using special geometric structures, and solutions of geometric equations, to analyse topology, especially in higher dimensions where there is such a wealth of structures to play with. It is here that physics has played an essential guiding role; in particular Yau recast one of the fundamental questions of physics – the unification of general relativity and quantum mechanics – as a important motivating problem in geometry, namely to find a geometric structure which resembles general relativity at large scales, and quantum physics at small scales; while it seems we still have not achieved this goal, the geometric constructions that have come out of attempts towards this goal (e.g. from string theory) have been proven to be extraordinarily useful in geometric topology. Yau also ventured the personal opinion that whatever the geometric structure is that will unify relativity and quantum mechanics, it is likely that plenty of bundles and cycles will be involved (and their presence will be felt particularly strongly in regions of classical singularity, such as black holes).

Yau then turned to the topic of how to actually build these geometric structures. Actually, there are two important questions here: (a) existence of these structures, and (b) understanding the moduli space of solutions. Here we shall focus primarily on (a). Ideally, one should find necessary and sufficient conditions for the existence of any given geometric structure in terms of algebraic topological data (e.g. homology classes, characteristic classes, or homotopy groups). In general this goal looks very difficult, but there are certainly some successes of this type. For instance, in low dimensions $n \leq 4$ it is understood which

2n-dimensional manifolds admit an almost complex structure (of n complex dimensions, of course); this problem is equivalent to lifting a map from the manifold to the classifying space BSO(2n) of SO(2n) to a map into the classifying space BU(n) of U(n). In principle one should be able to extend these sorts of results to higher dimensions and to other structure groups, but this has not yet been fully accomplished.

Yau made the interesting point that existence of geometric structures was often tied up (somewhat paradoxically) with uniqueness, or at least finite-dimensionality or compactness of the solution space. Generally speaking, it is only with this finite dimensionality that techniques from elliptic and parabolic PDE can be effective in locating solutions, because the smoothing properties of these PDE tend to restrict the solution space to be finite dimensional or compact. Unfortunately, there are important situations in which the space of geometric structures to be located is infinite dimensional, even after fixing all relevant topological data; for instance, a manifold can have an infinite-dimensional family of complex or symplectic structures with a fixed topology.

For general manifolds, one relies mainly on two major techniques to build geometric structures:

1. *Gluing techniques*, which builds structures on complicated manifolds by performing surgery on structures on simpler manifolds. This "naive" approach can lead to non-trivial results if used correctly, but often fails due to significant obstructions (such as those arising from convexity).
2. *Variational techniques*, which builds structures by trying to minimise some sort of energy-like functional. An important subclass of this technique uses a parabolic flow (such as a gradient flow for the energy functional) to obtain the minimiser.

In this lecture Yau focused only on the former; he will talk about the latter in the next lecture. In order to glue properly, one needs to carefully select the boundary of the region to cut out of both initial manifolds to create the glued manifold; this requires a good understanding of geometric structure on manifolds with boundary. A model example of this approach is Thurston's celebrated work on constructing hyperbolic metrics on various types of 3-manifolds (leading eventually to his proof of his geometrisation conjecture for Haken manifolds). Thurston used a version of the Mostow rigidity theorem for hyperbolic manifolds with boundary due to Ahlfors-Bers, Marden, and Sullivan, which asserted that such manifolds can be essentially determined from the conformal structure of the boundary and from the fundamental group. This essentially reduces the gluing problem to a lower dimensional problem, namely that of matching conformal structures on the two-dimensional boundaries, which he achieved by Teichmüller theory (which was discussed in the previous lecture), and in particular by introducing a fixed point theorem on the Teichmüller space.

Yau noted that one serious issue with gluing is that the boundary surface often "wants" to be "convex" on both sides, which is difficult to achieve unless the surface is chosen to be somehow "flat". This is a rather vague principle and depends on exactly what one is trying to glue together; for instance, for complex structures it is pseudoconvexity which is relevant rather than convexity, and "flatness" means, among other things, that the normal bundle of the boundary is trivial. Even then, there are further obstructions to gluing that are still not completely understood.

The work of Taubes showed that it was possible to combine gluing techniques with more analytical techniques, basically by first gluing together two geometric structures to form a singular "approximate" geometric structure, and then using PDE methods (more precisely, singular perturbation theory) to convert this singular structure back into a smooth and exact geometric structure. Using this strategy, Taubes showed how to glue instantons into a given four-dimensional manifold to create anti-self dual bundles on that manifold (or on that manifold with sufficiently many copies of $\mathbb{CP}^2$ attached); this work forms a fundamental component of Donaldson theory. (It is slightly unfortunate that one needs to add all these copies of $\mathbb{CP}^2$; this lets one evade some topological obstructions, but by the same token also destroys some of the topological information of the original manifold.) These anti-self dual bundles can lead to further interesting geometric objects, for instance by using a twistor construction of Penrose one can then generate large families of complex manifolds of three complex dimensions which fibre as sphere bundles over 4-manifolds.

Unfortunately, while the singular perturbation method is very powerful for demonstrating *existence* of geometric structures, it has so far proven to be fairly ineffective at understanding the *moduli space* of such structures; this latter question is apparently of particular interest in M-theory.

In general, these existence problems are well understood in low dimensions but are still very difficult in higher dimensions, for instance it is not known which manifolds admit affine structures (i.e. an atlas where all the coordinate transformations are affine maps). Yau mentioned a particularly simple problem which was still unsolved: is it true that a compact affine manifold must necessarily have vanishing Euler characteristic? This is known if the affine connection is complete.

One can study affine structures (or projective, or conformal) by the *developing map* of that structure, which maps the manifold to the universal cover ($\mathbb{R}^n$ for affine structures, $\mathbb{RP}^n$ for projective, and $S^n$ for conformal). In principle, this reduces matters to studying the action of discrete groups on a domain in this universal cover. Unfortunately, there is a major difficulty, namely that the developing map can be non-injective. (There is apparently exactly one general result in which injectivity can be established, namely that of conformally flat complete manifolds with positive curvature, by a result of Schoen and Yau – but this uses a lot of firepower to prove, in particular comparing various Green's functions to each other using the positive mass theorem.)

Yau then noted that our understanding of geometric structures improves considerably when one has supersymmetry; indeed, it seems that this concept is now more popular among geometers than among physicists, given that it has not yet been established that supersymmetry actually exists in our physical universe. In contrast, supersymmetry, or concepts inspired by supersymmetry, have proven to be rather fruitful in geometry; a good example is the Seiberg-Witten theory of 4-manifolds, which was inspired by supersymmetric Yang-Mills theory. (Apparently, when this theory was first introduced, it was not immediately appreciated, as the importance of supersymmetry to geometry was not fully established then.) One major application of this theory was by Taubes, who used the spinor-based Seiberg-Witten invariants to construct pseudoholomorphic curves in large classes of almost complex manifolds; in the special case of $\mathbb{CP}^2$, these curves are in fact genus zero and low degree, which in combination with the work of Kodiara and Hirzebruch can be used to conclude the remarkable fact that this manifold has no symplectic structure other than the obvious one. (The corresponding problem in higher dimensions is wide open, due to the lack of any counterpart to Taubes' result.)

[*Update*, May 18: Corrected the provenance of the Mostow rigidity theorem for hyperbolic manifolds with boundary.]

[*Update*, May 19: Finally figured out how to draw a commutative diagram.]

## Comments (7)

 **Richard** · 18 May 2007

Hello,
The "version of Mostow Rigidity" you mention, which states that the (minimally parabolic) geometrically finite hyperbolic structures on a manifold (with incompressible boundary) are parameterized by the Teichmüller space of the boundary, should be attributed to Ahlfors-Bers-Marden-Sullivan. Of course, the heart of Geometrization for Haken manifolds is the fixed point problem and its solution, and that's certainly Thurston's.
Cheers.

 **Anonymous** · 18 May 2007

Out of place, but congrats on your election to the Royal Society!

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 **At the AustMS conference « What's new** · 2 October 2009

[...] Mohammed Abuzaid gave a plenary lecture entitled "Functoriality in homological mirror symmetry". One thing I learned from this talk was that the (partially conjectural) phenomenon of (homological) mirror symmetry is one of several types of duality, in which the behaviour of maps into one mathematical object (e.g. immersed or embedded curves, surfaces, etc.) are closely tied to the behaviour of maps out of a dual mathematical object (e.g. functionals, vector fields, forms, sections, bundles, etc.). A familiar example of this is in linear algebra: by taking adjoints, a linear map into a vector space can be related to an adjoint linear map mapping out of the dual space . Here, the behaviour of curves in a two-dimensional symplectic manifold (or more generally, Lagrangian submanifolds in a higher-dimensional symplectic manifold), is tied to the behaviour of holomorphic sections on bundles over a dual algebraic variety, where the precise definition of "behaviour" is category-theoretic, involving some rather complicated gadgets such as the Fukaya category of a symplectic manifold. As with many other applications of category theory, it is not just the individual pairings between an object and its dual which are of interest, but also the relationships between these pairings, as formalised by various functors between categories (and natural transformations between functors). (One approach to mirror symmetry was discussed by Shing-Tung Yau at a distinguished lecture at UCLA, as transcribed in this previous post.) [...]

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